

POSTERIOR DISTRIBUTIONS OF MEASURES OF ASSOCIATION BETWEEN PREDICTORS OF HORSE RACING RESULTS

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Summary: The daily newspapers regularly publish the predictions of several predictors of forthcoming horse races in the country. These predictions play an important role among bettors who may believe more in one predictor than another. Questions arising are whether there are any real differences among these predictors, and how good their predictions are. In this paper a multivariate Bernoulli model is used to compare the predictions of three predictors namely Computaform, City Press and “Die Volksblad”. The posterior distributions of the pairwise measures of association between predictors are obtained through the Gibbs sampler. Only one type of betting is considered, namely the trifecta-box betting where a prediction of the first three horses in a race is considered. The box betting means that the three horses finishing in any permutation of the prediction is a success.

1. Introduction

Computaform is a company that supplies predictions on almost all horse races that are run in South Africa. They base their predictions on the past performances of horses only and are considered as an important source of information for any bettor. Almost all newspapers then supply, additional to

Computaform's predictions, their own predictions using their own expertise. These experts may use further information such as the jockey, the trainer of the horse, the distance of the race and some other information. We chose two newspapers namely City Press and "Die Volksblad" together with Computaform to analyse their predictions. In section 1 the predictions of only Computaform are considered and the probability of a correct prediction estimated. It is also shown that a bettor is bound to lose from these predictions only. In section two Computaform and City Press are compared. The analysis is done by defining the bivariate Bernoulli distribution and the Bayes estimates found through the Gibbs sampler. The odds-ratio is the parameter of interest and the posterior distribution of this measure of association follows as a byproduct from the Gibbs sampler. Altham (1969) gave an exact expression for the posterior probability of this measure being less than one, in other words for the association between two predictors to be negative. The posterior probability is derived assuming a Dirichlet prior on the parameters. She showed that this probability becomes Fisher's exact probability for testing no association in certain cases. The posterior distribution of the measure of association derived here is based on a Dirichlet prior obtained from a training sample. A comparison of the probability of negative association derived from this posterior with that obtained through Altham's results, shows very small probabilities. The results are extended to the three-variate Bernoulli distribution by considering the predictions of Computaform, City Press and "Die Volksblad" simultaneously and illustrates a method for analyzing more than three variables.

Lindley (1964) proposed a Bayesian analysis for contingency tables and mentioned the use of a prior obtained from data similar to that to be analyzed. In the approach followed here a Dirichlet prior is constructed by using a training sample with a uniform prior. A review of the work that has been done in the Bayesian approach can be found in Leonard and Hsu (1994). We followed the Bayesian approach here because of the insight one gets on the information of the measures of association by considering their posterior distributions.

2. Results of trifecta-box bettings on the predictions of Computaform

2.1 Results on bettings

In our local morning newspaper, "Die Volksblad" of February 13, 1996 one finds for instance the following information on the first two of nine races to be run at Newmarket race course during that afternoon:

Table 1: Predictions of the first four horses in two races by three predictors

Race	Time	"Die Volksblad"		Computaform		City Press	
1	12:45	6	Kal Sufi	6	Kal Sufi	11	Pumpkin Pie
		11	Pumpkin Pie	16	State Banquet	6	Kal Sufi
		16	State Banquet	11	Pumpkin Pie	16	State Banquet
		14	Seize Gold	2	Bacara's Girl	14	Seize Gold
2	13:15	4	Don Primo	4	Don Primo	3	Devlish Affair
		5	Great Show	5	Great Show	4	Don Primo
		8	Pol Roger	10	Somerset Man	5	Great Show
		10	Somerset Man	3	Devlish Affair	10	Somerset Man

They also supplied information on each horse such as its performance in the previous races, the trainer, the jockey, the weight it carries, the place it has drawn for a starting position and the distance of the race. This information will not be used.

The results of the races, together with the payouts are published in the newspapers the following day. Since we look only at the first three places of a race and the trifecta-box bettings, we are only interested in results such as shown in table 2.

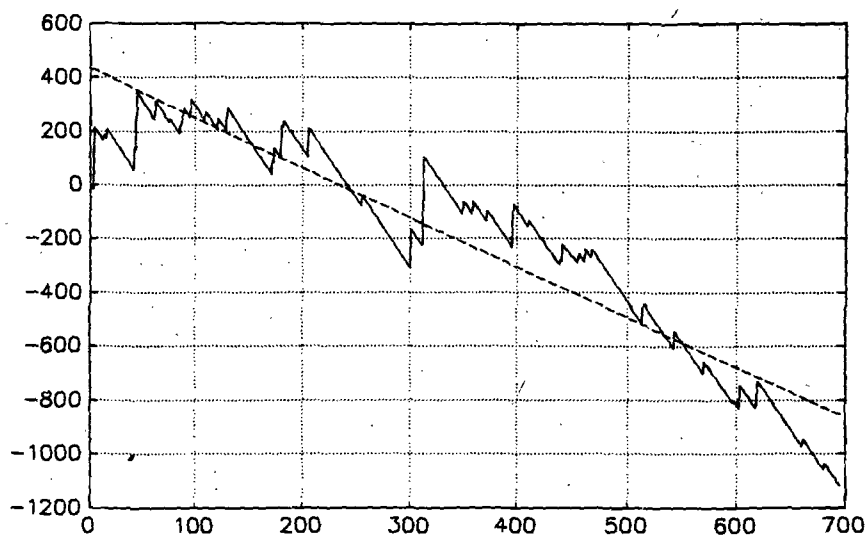
As can be seen from table 2, Computaform predicted only two races correctly on a trifecta-box bet, namely race 7 at Newmarket (3,4,9) with a payout of R54 and race 9 at Newmarket (10,11,7) with a payout of R57. A bet of R6 on each of the above 28 races would cost R168 and the total winnings would be $R54 + R57 = R111$. This means a loss of R57. Figure 1 shows the profits/losses on 697 races on trifecta-box bettings of Computaform predictions. It shows a loss and is certainly not a good investment.

2.2 Computaform's probability of success

We shall now look at the probability of a correct prediction of Computaform on the first three places if their prediction is boxed, that is any permutation of their prediction is taken as a successful outcome. If their prediction is for instance (3,4,9), then successful outcomes will be (3,4,9), (3,9,4), (4,3,9), (4,9,3), (9,3,4) and (9,4,3).

Table 2: Results of 28 races run on 17–18 October 1995 on three race courses, namely Milnerton (Cape), Greyville (Natal) and Newmarket (Gauteng)

Place	Race number	Computaform prediction			Result of race			Trifecta payout (R)
		1	2	3	1	2	3	
Milnerton	1	1	6	13	1	8	16	168
"	2	4	8	9	4	5	10	277
"	3	3	10	6	3	10	2	19
"	4	1	3	6	9	8	5	513
"	5	9	12	2	16	8	11	1409
"	6	8	4	1	7	14	1	604
"	7	2	1	9	1	6	10	1316
"	8	6	5	1	5	1	8	456
Greyville	1	7	12	4	6	15	12	815
"	2	2	1	7	1	13	5	34608
"	3	11	6	9	6	11	1	406
"	4	5	6	8	3	10	12	5785
"	5	3	2	4	3	11	4	915
"	6	1	4	2	9	2	4	833
"	7	2	5	8	2	8	7	68
"	8	7	6	3	3	5	7	122
"	9	7	5	8	5	7	2	33
"	10	7	3	9	7	11	9	7
Newmarket	1	7	16	20	3	17	6	1874
"	2	17	20	9	17	11	9	68
"	3	12	2	5	12	2	11	46
"	4	15	5	11	5	1	6	150
"	5	13	8	12	15	16	18	597
"	6	5	2	3	11	12	2	2789
"	7	4	3	9	3	4	9	54
"	8	4	6	9	4	6	1	192
"	9	7	11	10	10	11	7	57
"	10	4	7	10	13	4	16	253



Races

Figure 1. Winnings/losses on trifecta-box bettings of R6 per betting on Computaform's predictions with a linear trend.

The model is obviously a Bernoulli model with the random variable

$$X = \begin{cases} 1 & \text{if prediction is successful} \\ 0 & \text{if prediction is not successful.} \end{cases}$$

Let $p_x = P(X = 1)$, then

$$P(X = x) = p_x^x (1 - p_x)^{1-x}, \quad x = 0, 1.$$

The data consists of $n = 697$ races denoted by $x_1, x_2, \dots, x_n$, and the prior on p_x is taken to be uniform, i.e. $\pi(p_x) = 1$, $0 < p_x < 1$.

The posterior probability distribution of p_x is obviously

$$\pi(p_x | x_1, \dots, x_n) \propto p_x^{\sum x} (1 - p_x)^{n - \sum x}, \quad 0 < p_x < 1.$$

This is a Beta ($\alpha = \sum x + 1$, $\beta = n - \sum x + 1$) distribution. In our example $\sum x = 43$, so that the posterior mean of p_x is $\hat{p}_x = \frac{\sum x + 1}{n + 2} = \frac{44}{699} = 0.0629$. This is defined as a Bayes estimate.

Our estimate of the chance of a Computaform prediction on a trifecta-box betting being successful is therefore 6.29%.

It also means that one will be successful on the average of one out of $699/44 \cong 16$ bets.

3. Predictions of Computaform and City Press

We shall now consider the predictions of Computaform and City Press and examine the association between their predictions. The same trifecta-box betting is considered.

Let $X = \begin{cases} 1 & \text{Computaform's prediction is successful} \\ 0 & \text{Computaform's prediction is not successful} \end{cases}$

and $Y = \begin{cases} 1 & \text{City Press's prediction is successful} \\ 0 & \text{City Press's prediction is not successful} \end{cases}$

The joint probability distribution of X and Y , called the bivariate Bernoulli distribution, is given by

$$P(X = x, Y = y) = p_0^{1-x-y} p_x^x p_y^y \theta^{xy}, \quad x = 0, 1; y = 0, 1 \quad (1)$$

where

$$\theta = \frac{p_0 p_{xy}}{p_x p_y}, \quad p_0 = P(X = 0, Y = 0), \quad p_x = P(X = 1, Y = 0),$$

$$p_y = P(X = 0, Y = 1), \quad p_{xy} = P(X = 1, Y = 1) = 1 - p_0 - p_x - p_y.$$

θ is called the measure of association between X and Y and is the parameter of interest. It is obvious that if X and Y are independent, then $\theta = 1$.

θ is also known as the odds-ratio, much favoured by epidemiologists and others. " $\theta = 1$ " versus " $\theta < 1$ " can be tested by the Fisher exact test for 2×2 contingency tables. Altham (1969) gave an exact expression for the posterior probability of negative association, $P(\theta < 1 | \text{data})$ using a Dirichlet prior on p_x, p_y and p_{xy} .

A sample, which we will refer to as a training sample, of size $n = 168$ was drawn to get an idea of the parameters of the Dirichlet prior

$$\pi(p_0, p_x, p_y, p_{xy}) \propto p_0^{\alpha_0-1} p_x^{\alpha_x-1} p_y^{\alpha_y-1} p_{xy}^{\alpha_{xy}-1}$$

where $\alpha_0 = 1 + n - \Sigma x - \Sigma y$, $\alpha_x = 1 + \Sigma x - \Sigma xy$, $\alpha_y = 1 + \Sigma y - \Sigma xy$, $\alpha_{xy} = 1 + \Sigma xy$.

The training sample is obtained before the final sample is drawn, and the information gives prior estimates of the parameters. From the training sample we found; $\Sigma x = 7$, $\Sigma y = 6$, $\Sigma xy = 0$ and therefore the parameters are taken as $\alpha_0 = 156$, $\alpha_x = 8$, $\alpha_y = 7$, $\alpha_{xy} = 1$ if we start with a uniform prior and use the posterior as our prior. A sample of $n = 305$ races was then drawn with $\Sigma x = 15$, $\Sigma y = 15$, $\Sigma xy = 7$. The likelihood function for p_0 , p_x and p_y is given by

$$\ell(p_0, p_x, p_y | n, \Sigma x, \Sigma y, \Sigma xy) \propto p_0^{n - \Sigma x - \Sigma y} p_x^{\Sigma x} p_y^{\Sigma y} \theta^{\Sigma xy}.$$

The posterior now becomes a Dirichlet with parameters

$$\alpha'_0 = \alpha_0 + n - \Sigma x - \Sigma y + \Sigma xy = 438$$

$$\alpha'_x = \alpha_x + \Sigma x - \Sigma xy = 16$$

$$\alpha'_y = \alpha_y + \Sigma y - \Sigma xy = 15$$

$$\alpha'_{xy} = \alpha_{xy} + \Sigma xy = 8$$

Note that the information from the training sample could just as well be included in the sample and the posterior will be the same under a uniform prior.

The Bayes estimates of p_0 , p_x , p_y and p_{xy} obtained as the posterior means, are

$$E(p_0|n, \Sigma_x, \Sigma_y, \Sigma_{xy}) = \frac{\alpha'_0}{\alpha'_0 + \alpha'_x + \alpha'_y + \alpha'_{xy}} = 0.9182$$

$$E(p_x|n, \Sigma_x, \Sigma_y, \Sigma_{xy}) = \frac{\alpha'_x}{\alpha'_0 + \alpha'_x + \alpha'_y + \alpha'_{xy}} = 0.0335$$

$$E(p_y|n, \Sigma_x, \Sigma_y, \Sigma_{xy}) = \frac{\alpha'_y}{\alpha'_0 + \alpha'_x + \alpha'_y + \alpha'_{xy}} = 0.0314$$

$$E(p_{xy}|n, \Sigma_x, \Sigma_y, \Sigma_{xy}) = \frac{\alpha'_{xy}}{\alpha'_0 + \alpha'_x + \alpha'_y + \alpha'_{xy}} = 0.0168$$

The posterior mean and variance of θ are given by

$$E(\theta|n, \Sigma_x, \Sigma_y, \Sigma_{xy}) = \frac{\alpha'_0 \alpha'_{xy}}{(\alpha'_x - 1)(\alpha'_y - 1)} = 16.6857$$

$$\begin{aligned} \text{Var}(\theta|n, \Sigma_x, \Sigma_y, \Sigma_{xy}) &= \frac{(\alpha'_0 + 1)\alpha'_0(\alpha'_{xy} + 1)\alpha'_{xy}}{(\alpha'_x - 1)(\alpha'_x - 2)(\alpha'_y - 1)(\alpha'_y - 2)} \\ &\quad - 16.6857^2 \\ &= 83.8174 \end{aligned}$$

The posterior distribution of θ is simulated through the Gibbs sampler and a histogram from 15000 simulated values is shown in figure 2.

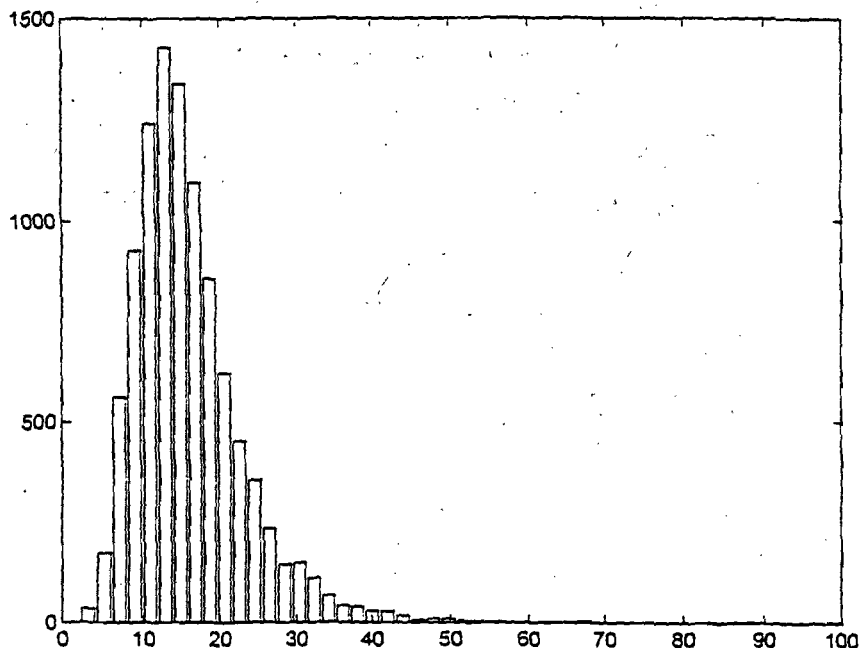


Figure 2: Posterior density of the measure of association $\theta = p_0 p_{xy} / p_x p_y$ between the successes of predictions by Computaform and City Press

It is clear that θ is much larger than one, meaning that there is a strong association between the predictions of Computaform and of City Press. None of the 15 000 simulated values of theta was less than or equal to one. An

estimate of $P(\theta \leq 1 | \text{data})$ is therefore smaller than 10^{-5} . The exact probability given by Altham is

$$\begin{aligned}
 & P(\theta < 1 | \Sigma x = 15, \Sigma y = 15, \Sigma xy = 7, \alpha_0 = 156, \\
 & \quad \alpha_x = 8, \alpha_y = 7, \alpha_{xy} = 1) \\
 &= \sum_{s=\max(v_{21}-v_{12}, 0)}^{v_{21}-1} \binom{v_{11}-1}{s} \binom{v_{22}-1}{v_{21}-1-s} / \binom{v_{..}-2}{v_{11}-1} \\
 &= 1.5292 \times 10^{-15}
 \end{aligned}$$

where

$$v_{ij} = n_{ij} + v'_{ij}$$

$$n_{11} = 305 - \Sigma x - \Sigma y + \Sigma xy = 268$$

$$n_{12} = \Sigma x - \Sigma xy = 8$$

$$n_{21} = \Sigma y - \Sigma xy = 8$$

$$n_{22} = \Sigma xy = 7$$

$$v'_{11} = \alpha_0, v'_{12} = \alpha_x, v'_{21} = \alpha_y, v'_{22} = \alpha_{xy}.$$

If $v'_{ij} = 1$, $1 \leq i, j \leq 2$, then Altham's probability becomes much smaller. The probability is then also Fisher's exact probability for testing $\theta = 1$ against $\theta < 1$ in a 2×2 table.

This shows that there is a very small chance of θ being one or less than one and therefore we can conclude that the predictions of Computaform and City Press show some association as $E(\theta | \text{data}) = 16.69$ indicates. If the case

$\Sigma xy = 15$ is considered, then there is high association. The expected value $E(\theta|\text{data})$ then becomes large – in the order of 10^4 . It is interesting to note that the maximum likelihood estimate for θ from the sample is $\hat{\theta} = \frac{282 \times 7}{8 \times 8} = 30.84$ and the mean of the simulated values shown in figure 2 is 16.07.

Similar results have been obtained for a comparison between Computaform and The Volksblad and City Press and “Die Volksblad”.

We shall now extend the analysis to three predictors Computaform, City Press and “Die Volksblad” on a trifecta-box setting.

4. Predictions of Computaform, City Press and “Die Volksblad”

$$\text{Let } X = \begin{cases} 1 & \text{Computaform successful} \\ 0 & \text{Computaform unsuccessful} \end{cases}$$

$$Y = \begin{cases} 1 & \text{City Press successful} \\ 0 & \text{City Press unsuccessful} \end{cases}$$

$$Z = \begin{cases} 1 & \text{“Die Volksblad” successful} \\ 0 & \text{“Die Volksblad” unsuccessful} \end{cases}$$

The three-variate Bernoulli distribution to describe the behaviour of the three variables is

$$P(X = x, Y = y, Z = z) = p_0^{1-x-y-z} p_x^x p_y^y p_z^z \theta_{xy}^{xy} \theta_{xz}^{xz} \theta_{yz}^{yz} \theta_{xyz}^{xyz},$$

$$x = 0, 1; y = 0, 1; z = 0, 1$$

where

$$\theta_{xy} = \frac{p_0 p_{xy}}{p_x p_y}, \quad \theta_{xz} = \frac{p_0 p_{xz}}{p_x p_z}, \quad \theta_{yz} = \frac{p_0 p_{yz}}{p_y p_z},$$

$$\theta_{xyz} = \frac{p_x p_y p_z p_{xyz}}{p_{xy} p_{xz} p_{yz} p_0}, \quad p_{xyz} = 1 - p_0 - p_x - p_y - p_z - p_{xy} - p_{xz} - p_{yz}$$

and

$$p_{xz} = P(X = 1, Y = 0, Z = 1) \text{ for instance.}$$

The data on $n = 305$ results are

$$\Sigma x = 15, \quad \Sigma y = 15, \quad \Sigma z = 14, \quad \Sigma xy = 7, \quad \Sigma xz = 6, \quad \Sigma yz = 5, \quad \Sigma xyz = 3.$$

We again used a training sample to find the following prior distribution on $p_0, p_x, p_y, p_z, p_{xy}, p_{xz}, p_{yz}$ and p_{xyz} namely a Dirichlet

$$(\alpha_0, \alpha_x, \alpha_y, \alpha_z, \alpha_{xy}, \alpha_{xz}, \alpha_{yz}, \alpha_{xyz})$$

$$\begin{aligned} \text{with } \alpha_0 &= 151, & \alpha_x &= 7, & \alpha_y &= 4, & \alpha_z &= 5, \\ \alpha_{xy} &= 3, & \alpha_{xz} &= 3, & \alpha_{yz} &= 2, & \alpha_{xyz} &= 1. \end{aligned}$$

The posterior becomes a Dirichlet $(\alpha'_0, \alpha'_x, \alpha'_y, \alpha'_{xy}, \alpha'_{xz}, \alpha'_{yz}, \alpha'_{xyz})$

where

$$\begin{aligned} \alpha'_0 &= 427, & \alpha'_x &= 12, & \alpha'_y &= 10, & \alpha'_z &= 11, \\ \alpha'_{xy} &= 7, & \alpha'_{xz} &= 6, & \alpha'_{yz} &= 4, & \alpha'_{xyz} &= 4. \end{aligned}$$

where

$$\alpha'_0 = \alpha_0 + n - \Sigma x - \Sigma y - \Sigma z + \Sigma xy + \Sigma xz + \Sigma yz - \Sigma xyz$$

$$\alpha'_x = \alpha_x + \Sigma x - \Sigma xy - \Sigma xz + \Sigma xyz$$

$$\alpha'_y = \alpha_y + \Sigma y - \Sigma xy - \Sigma yz + \Sigma xyz$$

$$\alpha'_z = \alpha_z + \Sigma z - \Sigma xz - \Sigma yz + \Sigma xyz$$

$$\alpha'_{xy} = \alpha_{xy} + \Sigma xy - \Sigma xyz$$

$$\alpha'_{xz} = \alpha_{xz} + \Sigma xz - \Sigma xyz$$

$$\alpha'_{yz} = \alpha_{yz} + \Sigma yz - \Sigma xyz$$

$$\alpha'_{xyz} = \alpha_{xyz} + \Sigma xyz$$

The expected values of the measures θ_{xy} , θ_{xz} , θ_{yz} can be calculated similarly to that for θ in the previous section, namely

$$E(\theta_{xy}|\text{data}) = \frac{\alpha'_0 \alpha'_{xy}}{(\alpha'_x - 1)(\alpha'_y - 1)} = 30.1919$$

$$E(\theta_{xz}|\text{data}) = 23.2909$$

$$E(\theta_{yz}|\text{data}) = 18.9778$$

The expected value of θ_{xyz} is

$$E(\theta_{xyz}|\text{data}) = \frac{\alpha'_x \alpha'_y \alpha'_z \alpha'_{xyz}}{(\alpha'_{xy} - 1)(\alpha'_{xz} - 1)(\alpha'_{yz} - 1)(\alpha'_0 - 1)} = 0.1033$$

The posterior distributions of these measures were simulated through the Gibbs sampler and are shown in figure 3. The means and variances of the simulated values correspond closely to the exact values. To get an idea of the meaning of $E(\theta_{xyz}|\text{data}) = 0.1033$ as a measure of association between the three predictors, we simulated the posterior of θ_{xyz} with Σ_{xyz} changed to the minimum value it can assume, namely $\Sigma_{xyz} = 0$ and the maximum value it can assume, namely $\Sigma_{xyz} = 5$. The posterior mean then ranges from 0.0028 to 2.5252. To attach these values to probabilities with respect to the posterior distribution of θ_{xyz} obtained from the data, we found that

$$P(\theta_{xyz} < 0.0028|\text{data}) = 0.0011$$

$$\text{and } P(\theta_{xyz} > 2.5252|\text{data}) = 10^{-3}.$$

These probabilities indicate small chances for no association and high association. We can therefore conclude that the three predictors are jointly fairly strongly associated.

An estimate of $P(X = 1) = p_x + p_{xy} + p_{xz} + p_{xyz}$ from the Gibbs simulations is 0.0602 compared to the true $(\alpha'_x + \alpha'_{xy} + \alpha'_{xz} + \alpha'_{xyz}) / \Sigma \alpha' = 0.0603$. This probability compares well with the one calculated in section 2 which is based on a different sample. The corresponding probability for City Press is 0.0519 and the same for "Die Volksblad".

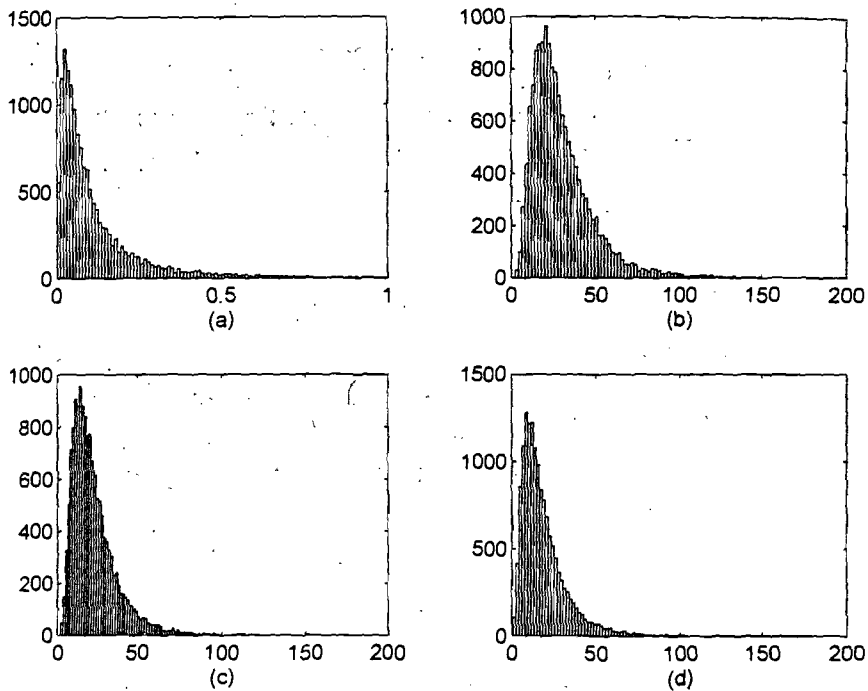


Figure 3. Posterior distributions of
 (a) θ_{xyz} , (b) θ_{xy} , (c) θ_{xz} , (d) θ_{yz} .

The distributions of θ_{xy} , θ and θ_{yz} show the posteriors of the measure of association pairwise, given that the third predictor is wrong. The posterior means of these measures are

$$E(\theta_{xy}|\text{data}) = 30.1919, \quad E(\theta_{xz}|\text{data}) = 23.2909 \quad \text{and} \quad E(\theta_{yz}|\text{data}) = 18.9778.$$

The posterior distributions show very small probabilities for the measure to be one or less than one. This means that conditional on one predictor being wrong, the other two show association.

5. Concluding remarks

The analysis shows a low probability for any of the three predictors Computaform, City Press and "Die Volksblad" to make a successful prediction on a trifecta-box betting. It shows association between these predictions pairwise, although very little between the three jointly. It would therefore be wise to look at the predictions of the three together. It is also true that the payouts on successful bets from Computaform's predictions are in general low because many bettors use these predictions.

6. Acknowledgement

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